Chapter 11: The Aggregate Expenditures Model

Practice Problems

Warm-Up: Quick Concept and Calculation Checks

a. In the aggregate expenditures model, prices are assumed to be sticky in the short run. When total spending changes, does that show up mainly as a change in *real output* or a change in *prices*?

b. True or False: equilibrium GDP always equals potential (full-employment) GDP.

c. At a GDP level where planned spending (AE) is *greater* than actual output, what happens to unplanned inventories, and which direction does that push future production?

d. At a GDP level where planned spending (AE) is *less* than actual output, what happens to unplanned inventories, and which direction does that push future production?

e. If the MPC in an economy is 0.75, what is the simple multiplier?

f. A country's aggregate expenditures at full-employment GDP fall $15 billion short of what is needed to reach full-employment GDP. Is this a recessionary or an inflationary expenditure gap?

Problem 1: Building the AE Schedule and Finding Equilibrium

A simple, private, closed economy (no government, no trade) has the consumption schedule $C = 150 + 0.75 \times GDP$ (in billions of dollars), and planned investment is constant at $I_g = 50$ billion regardless of the level of GDP.

Real GDP	Consumption (C)	Investment (I_g)	Aggregate Expenditures (AE)
700	675	50	?
800	750	50	?
900	825	50	?
1000	900	50	?

a. Fill in the Aggregate Expenditures column ($AE = C + I_g$) for each row.

b. At which GDP level in the table does AE equal GDP? This is equilibrium GDP.

c. At GDP $= 1000$, is AE greater than or less than GDP? What does this imply is happening to unplanned inventories and future production at that GDP level?

d. What is the MPC implied by this consumption schedule, and what is the multiplier?

Problem 2: Adding Government Spending and Taxes

Starting from the private economy in Problem 1 (equilibrium GDP $= \$800$ billion, MPC $= 0.75$), the government now adds $G = \$50$ billion in purchases.

a. By how much does the AE schedule shift at every GDP level?

b. Using the multiplier from Problem 1(d), by how much does equilibrium GDP change? What is the new equilibrium GDP?

c. The government now also imposes a lump-sum tax of $T = \$100$ billion. Since $MPC = 0.75$, consumption falls by $MPC \times T = 0.75 \times 100 = \$75$ billion at every GDP level. Using the multiplier, what is the tax's effect on equilibrium GDP by itself?

d. Combining the effects of G and T from parts (b) and (c), what is the new equilibrium GDP?

Problem 3: Net Exports and a Recessionary Gap

Continuing the same economy from Problem 2 (with both $G = \$50$ billion and $T = \$100$ billion in place, equilibrium GDP $= \$700$ billion), the economy now also has net exports of $X_n = \$50$ billion.

a. By how much does equilibrium GDP change due to net exports? What is the new equilibrium GDP?

b. Suppose this economy's full-employment (potential) GDP is $\$1{,}100$ billion. Is there a recessionary or an inflationary expenditure gap?

c. At full-employment GDP ($1{,}100$), aggregate expenditures are only $\$1{,}050$ billion. What is the size of the expenditure gap (the vertical shortfall between AE and the 45-degree line at full employment)?

d. Using the multiplier, if the government wanted to close this gap entirely, by how much would it need to increase spending?